VHR image (518 × 518)



Reference mask



urban agric. range forest

Frozen backbone

DINOv2 ViT-S/B/L
(+ ResNet-50 check)
no weights updated

Patch features

37 × 37 grid
768-d (ViT-B)
extracted once

Annotation regime

Dense: n fully annotated images
(n = 1 ... all)
Sparse: P random single-pixel points
(P = 100 ... 30,000)

Classifiers

Linear probe
Class-balanced probe
Cluster-then-label (K = 50)
Zero-label k-means

Baseline:
DeepLabV3 fine-tuning

Exact scoring

patch label → its 196 px
pixel-level confusion
per-class IoU, mIoU
patch-oracle ceiling
paired tests (Holm)

Protocol

DeepGlobe 5-fold CV × 3 draws · LoveDA train→val × 5 draws
urban↔rural transfer · λ, K sensitivity · 4 backbones